MATLAB Simulation Exercise

Statistical Signal Processing II

Detection of Random Signals

Group 09

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1 Theoretical Background

1.1 Energy Detector

In many signal detection problems, the objective is to determine whether a signal is present in noisy observations. In this laboratory exercise, we consider the detection of a Gaussian random signal embedded in white Gaussian noise (WGN). The received samples are modeled as a sequence of observations

$$x[n], \quad n = 0, 1, \dots, N-1$$

Our detection problem can thus be formulated as a binary hypothesis test

$$H_0 : x[n] = w[n] \tag{1}$$
$$H_1 : x[n] = s[n] + w[n] \tag{2}$$

where $s[n]$ represents the signal and $w[n]$ additive white Gaussian noise with variance σ^2. The noise is assumed independent of the signal. Additionally, the signal $s[n]$ is assumed to be a zero-mean, white, wide-sense stationary Gaussian random process with variance σ_s^2. Therefore;

$$s[n] \sim \mathcal{N}(0, \sigma_s^2)$$

and the noise samples follow

$$w[n] \sim \mathcal{N}(0, \sigma^2)$$

The observation vector is provided as

$$\mathbf{x} = [x[0], x[1], \dots, x[N-1]]^T$$

and will have different statistical distributions under the two hypotheses as shown below;

$$H_0 : \mathbf{x} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}) \tag{3}$$
$$H_1 : \mathbf{x} \sim \mathcal{N}(0, (\sigma_s^2 + \sigma^2)\mathbf{I}) \tag{4}$$

To determine which hypothesis is true, the Neyman Pearson (NP) detector is employed. The NP test compares the likelihood ratio to a decision threshold γ as shown below,

$$L(\mathbf{x}) = \frac{p(\mathbf{x}; H_1)}{p(\mathbf{x}; H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

Using the gaussian probability density functions for the two hypotheses, the likelihood ration can be simplified as [1];

$$\ell(\mathbf{x}) = \frac{N}{2} \ln\left(\frac{\sigma^2}{\sigma_s^2 + \sigma^2}\right) + \frac{\sigma_s^2}{2\sigma^2(\sigma_s^2 + \sigma^2)} \sum_{n=0}^{N-1} x^2[n]$$

Since the first term is constant, the decision rule simplifies to a comparison involving the energy of the received signal;

$$T(x) = \sum_{n=0}^{N-1} x^2[n]$$

The detector consequently decides;

$$T(x) \underset{H_0}{\overset{H_1}{\gtrless}} \gamma'$$

The test statistic represents the total energy of the observed samples, and the resulting detector is known as an energy detector [1]. That is, when the signal is present, the received energy increases due to the additional signal power. The distribution of the test statistic depends on the hypothesis. As a result, $T(x)$ is the sum of squares of Gaussian random variables and thus it follows a chi–square distribution as shown below;

$$\frac{T(x)}{\sigma^2} \sim \chi_N^2 \quad \text{under } H_0 \tag{5}$$

$$\frac{T(x)}{\sigma_s^2 + \sigma^2} \sim \chi_N^2 \quad \text{under } H_1 \tag{6}$$

Using these distributions, the probability of false alarm and probability of detection can be written as

$$P_{FA} = P(T(x) > \gamma' | H_0) \tag{7}$$

$$= Q\left(\frac{\gamma'}{\sigma^2}\right) \tag{8}$$

$$P_D = P(T(x) > \gamma' | H_1) \tag{9}$$

$$= Q\left(\frac{\gamma'}{\sigma_s^2 + \sigma^2}\right) \tag{10}$$

The detection performance depends strongly on the signal-to-noise ratio (SNR), defined as

$$\mathrm{SNR} = \frac{\sigma_s^2}{\sigma^2}$$

As the SNR increases, the distributions under H_0 and H_1 become more separated, which improves the probability of detection for a fixed false alarm rate [1].

1.2 General Linear Model

In the general linear model, the detection problem observation vector $\mathrm{x} \in \mathrm{R}^N$ is described by the following binary hypothesis test [1],

$$H_0 : \mathrm{x} = \mathrm{w} \tag{11}$$

$$H_1 : \mathrm{x} = \mathrm{Hd} + \mathrm{w} \tag{12}$$

where H is an $N \times M$ known matrix representing the channel or linear system, d is an $M \times 1$ random data vector, and w is additive white Gaussian noise. The statistical assumptions are

$$\mathrm{d} \sim \mathcal{N}(0, \sigma_s^2 \mathrm{I}), \qquad \mathrm{w} \sim \mathcal{N}(0, \sigma^2 \mathrm{I})$$

with d and w mutually independent. Letting the signal component be defined as

$$\mathrm{s} = \mathrm{Hd} \tag{13}$$

The covariance matrix of the signal is therefore [1];

$$\mathrm{C}_s = E[\mathbf{ss}^T] \tag{14}$$

$$= E[\mathrm{Hdd}^T\mathrm{H}^T] \tag{15}$$

$$= \mathrm{H}E[\mathrm{dd}^T]\mathrm{H}^T \tag{16}$$

$$= \sigma_s^2 \mathrm{HH}^T \tag{17}$$

Thus, for the signal present hypothesis, the observation vector follows as;

$$\mathrm{x} \sim \mathcal{N}(0, \mathrm{C}_s + \sigma^2 \mathrm{I})$$

while under H_0

$$\mathrm{x} \sim \mathcal{N}(0, \sigma^2 \mathrm{I})$$

Applying the Neyman–Pearson likelihood ratio test leads to the following test statistic [1];

$$T(\mathrm{x}) = \mathrm{x}^T \mathrm{C}_s(\mathrm{C}_s + \sigma^2 \mathrm{I})^{-1}\mathrm{x}$$

Therefore, the detector decides

$$T(\mathrm{x}) \underset{H_0}{\overset{H_1}{\gtrless}} \gamma$$

where γ is a threshold chosen to satisfy a desired probability of false alarm P_{FA}. This detector is known as the estimator–correlator since the signal component is estimated from the observation using a Wiener filter first [1],

$$\hat{\mathrm{s}} = \mathrm{C}_s(\mathrm{C}_s + \sigma^2 \mathrm{I})^{-1}\mathrm{x}$$

and then the estimate is correlated with the received data.

2 Results and Discussions

2.1 Energy Detector

Figure 1, below shows the probability of detection P_D as a function of signal to noise ratio (SNR) for different values of probability of false alarm P_{FA} with $N = 25$ samples. The probability of detection increases monotonically with SNR for all values of P_{FA}. At very low SNR values, the signal power is much smaller in comparison to the noise power, making it difficult for the detector to distinguish between the hypotheses H_0 and H_1. As the SNR increases, the signal component becomes more dominant and the detector is able to identify the presence of the signal with higher accuracy.

The effect of the probability of false alarm on the detection performance can be seen as larger values of P_{FA} correspond to lower detection thresholds, which increases the likelihood that the detector declares the presence of a signal. Consequently, higher P_D values are achieved at lower SNR levels when P_{FA} is large. For example, when $P_{FA} = 10^{-1}$, the probability of detection begins to increase significantly at relatively low SNR values. In contrast, smaller values of P_{FA} enforce strict thresholds to limit false alarms, resulting in curves that shift to the right and require higher SNR to achieve the same detection probability.

Overall, the simulation results follow the expected theoretical behavior of energy detection. Increasing the SNR improves the separation between the hypotheses H_0 and H_1, leading to higher detection probabilities. At the same time, reducing the false alarm probability makes the detector more

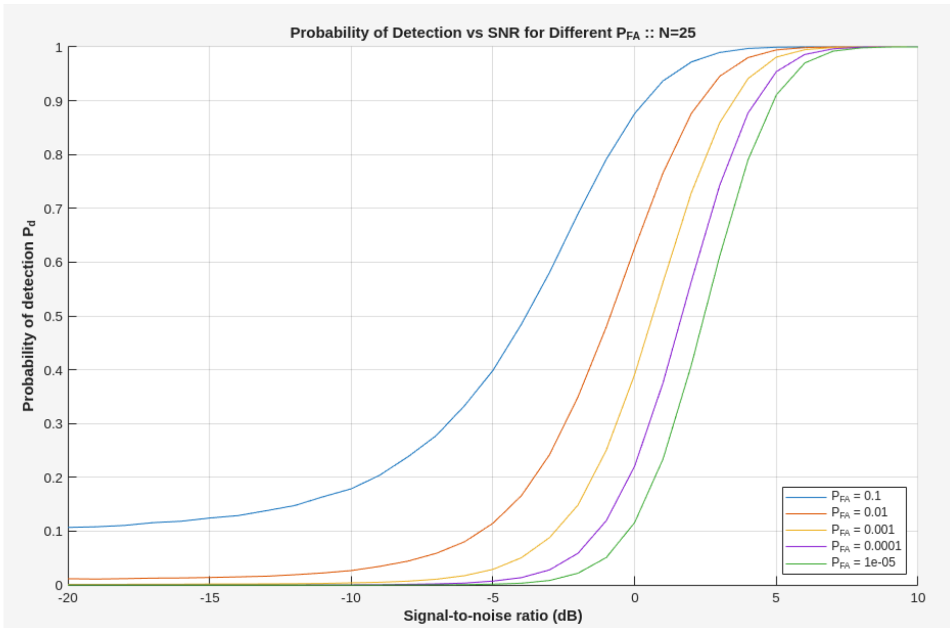

Figure 1: Probability of Detection with SNR for N=25

conservative, which requires stronger signal conditions to achieve reliable detection. The obtained curves closely resemble the theoretical detection performance presented in Example 5.1 [1], confirming that the implemented simulation correctly reproduces the expected behavior of the energy detector.

Figure 2 below, shows the probability of detection P_D as a function of the SNR for different values of probability of false alarm P_{FA} when the number of samples is increased to $N = 50$. As expected, the probability of detection increases with SNR for all values of P_{FA}. Compared to the case with $N = 25$, the curves become steeper and shift slightly to the left, indicating improved detection performance. As the number of samples N increases, the separation between the distributions under H_0 and H_1 becomes larger. At the same time, the relative variability of the test statistic decreases due to averaging over more observations, making the detector more stable.

The influence of P_{FA} remains consistent with the previous case. Larger values of P_{FA} result in lower detection thresholds and therefore achieve higher detection probabilities at lower SNR values. Conversely, smaller values of P_{FA} impose stricter thresholds, which reduces false alarms but requires higher SNR to achieve similar detection performance. Overall, increasing the

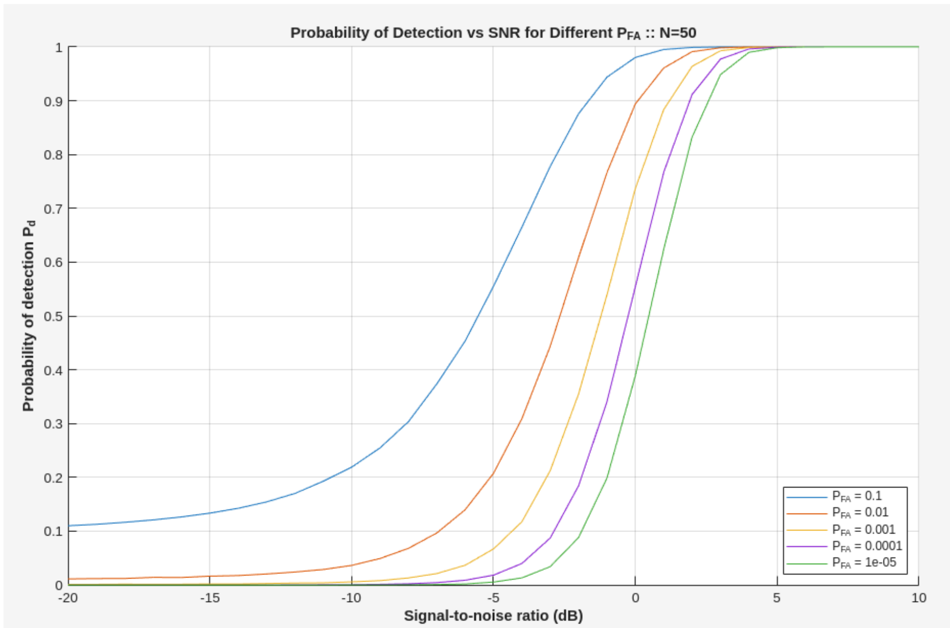

Figure 2: Probability of Detection with SNR for N=50

number of samples improves the detector's sensitivity and leads to more reliable signal detection.

2.2 Estimator-Correlator Detector

In Task 2, we extend the detection problem to the general case of the linear model. Instead of assuming a white signal, the signal is generated through a linear transformation as shown in equation 13. In this experiment the parameters were chosen as $N = 50$ and $M = 2$. Unlike Task 1, the signal samples are now correlated due to the presence of the matrix H. Consequently, the estimator-correlator detector is implemented using the quadratic test statistic shown below;

$$T(x) = x^T C_s (C_s + \sigma^2 I)^{-1} x,$$

where the signal covariance matrix is given by

$$C_s = \sigma_s^2 H H^T.$$

Figure 3, shows the probability of detection as a function of SNR for different false-alarm probabilities. Probability of detection increases smoothly as the SNR increases. At low SNR values, noise dominates the observation

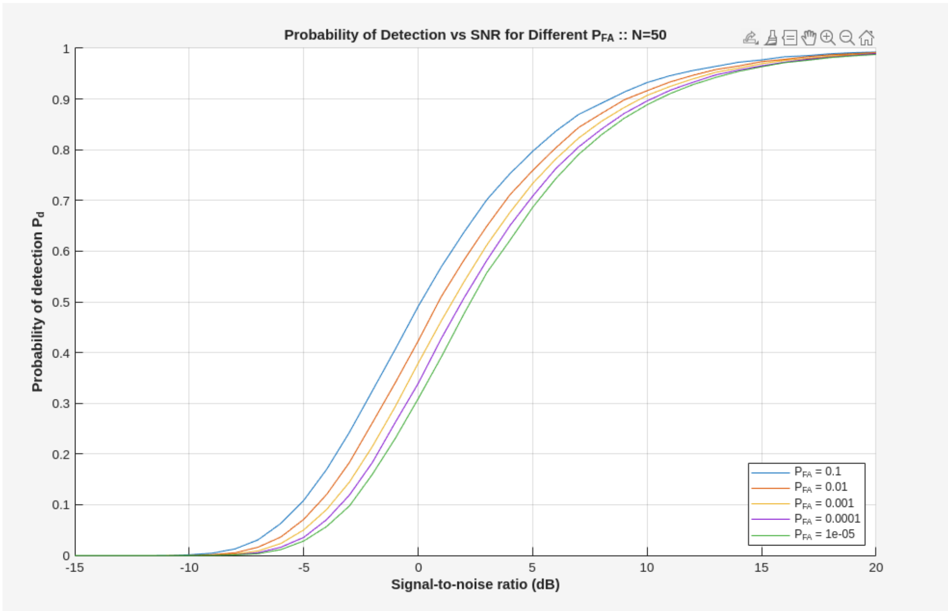

Figure 3: Figure 3: Probability of Detection with SNR for N=50, M=2

and the detector has difficulty distinguishing between the two hypotheses. As the SNR increases, the signal becomes stronger relative to the noise, which makes it easier for the detector to identify the presence of the signal. This behavior is consistent for all values of P_{FA}, although smaller values of P_{FA} require slightly higher SNR to achieve the same detection performance. It can also be observed that the curves corresponding to different values of P_{FA} are closer to each other compared to the energy detector case.

Compared to the energy detector used in Task 1, the estimator-correlator explicitly exploits the structure of the signal through the matrix H. Instead of simply measuring the total received energy, the detector weights the observation vector according to the signal covariance. Since the signal occupies only a lower-dimensional subspace ($M = 2$), the detector can better distinguish signal components from noise. As a result, the detection performance improves more smoothly with SNR, reflecting the influence of the signal structure introduced by the matrix H. This also makes the detection performance less sensitive to the choice of P_{FA}, which explains why the curves appear closer together.

3 Conclusion

In this work, we investigated the performance of signal detection in additive white Gaussian noise using two approaches i.e. energy detector and estimator-correlator detector. The energy detector was first analyzed for different values of the probability of false alarm and different observation lengths. Our results show that the probability of detection increases with the signal to noise ratio. Additionally, larger values of P_{FA} allow higher detection probabilities at lower SNR values, while smaller values require stricter thresholds and therefore need higher SNR for reliable detection. Increasing the number of samples from $N = 25$ to $N = 50$ improved the detector performance by increasing the separation between the distributions under H_0 and H_1 and reducing the variability of the test statistic.

Furthermore, we extended the detection problem to the general linear model, where the signal is generated through a linear transformation i.e. s = Hd. Compared to the energy detector, the estimator-correlator utilizes additional information about the signal through the matrix H, which improves the ability of the detector to distinguish the signal from noise. The results showed a smooth improvement of detection probability with increasing SNR and less sensitivity to the choice of P_{FA}. Overall, the simulations confirmed the theoretical behavior of both detectors and demonstrated the advantages of exploiting signal structure in statistical signal detection.

References

[1] S. M. Kay, *Fundamentals of Statistical Signal Processing: Detection Theory, vol. 2.* Prentice Hall International, 1998.